AMD: AM9080 on FPGA and DOOM Application

Statement and Scope of the Problem(s)

Context

The AMD Am9080 is a historically significant microprocessor, representing AMD's early role as a second-source provider for the Intel 8080 architecture. While the original Am9080 is limited by the technology of its era, it provides an excellent platform for exploring fundamental processor design, digital systems integration, and hardware/software co-design.

This project proposes that students recreate an **Am9080-compatible CPU using FPGA technology**, and design a supporting system with modern peripheral enhancements. Rather than attempting to run an unmodified version of the original DOOM engine, the students will demonstrate the capabilities and limits of the recreated platform by developing a **simplified, DOOM-inspired graphical application** suitable for an 8080-class processor with extended memory and peripherals.

Senior Design Focus

- RTL design and functional verification of a classic microprocessor
- FPGA-based CPU implementation with pin-compatible interfaces
- Motherboard, peripheral, and memory-expansion design (PCB)
- Hardware/software co-design and embedded systems development

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Overall Project Goal(s)

Context

The overall goal of this project is to recreate the functionality of the AMD Am9080 CPU and surrounding system architecture in a modern, educationally valuable form. By combining a historically accurate instruction-set implementation with carefully scoped modern enhancements, students will explore both the original design constraints of early microprocessors and the impact of architectural extensions.

Senior Design Focus

Recreate the AMD Am9080 instruction set using FPGA technology and design an accompanying motherboard and peripherals (video, memory expansion, input/output) that enable a playable, DOOM-inspired graphical demonstration while remaining generally compatible with original Am9080 software.

Project Objectives

1. **Pin-compatible FPGA implementation of the AMD Am9080**
   - Implement the full Am9080 instruction set with functional correctness.
   - Provide a pin-compatible interface such that the FPGA implementation can serve as a drop-in replacement (or equivalent interposer-based replacement) for an original Am9080.
   - Include a selectable **"Turbo Mode"** that allows the FPGA clock to run at higher-than-original frequencies for experimental performance evaluation.
2. **Motherboard and system platform**

   Design or adapt a motherboard (open-source solutions allowed) to support:

   - FPGA-based Am9080 CPU
   - RAM
   - Non-volatile storage
   - Peripheral input/output (keyboard, mouse, or equivalent)
   - Display output suitable for low-resolution graphics
3. **Memory expansion**
   - Implement at least **4 MB of pageable memory** beyond the 64 KB native addressing capability of the Am9080.
   - Memory expansion may be integrated directly on the motherboard or provided via a memory-only daughterboard.
   - Addressing extensions may use bank switching or I/O-mapped mechanisms while preserving Am9080 software compatibility.
4. **DOOM-inspired software demonstration**

   Develop a simplified, playable graphical application inspired by DOOM-style gameplay.

   The demonstration may use:

   - Grayscale rendering
   - Reduced resolution
   - Fixed-point arithmetic
   - Simplified rendering techniques (e.g., Wolfenstein-style raycasting)

     The software should demonstrate a minimal but complete gameplay loop (movement, environment rendering, interaction).

Expected Project Deliverables

Hardware Deliverables

1. **FPGA Am9080 implementation**
    - FPGA design implementing the full Am9080 instruction set.
    - Pin-compatible interface enabling socket or interposer use.
    - Configurable turbo-mode operation.
2. **System motherboard**
    - CPU socket or FPGA interface
    - RAM
    - Storage
    - Peripheral ports (keyboard and basic input)
    - Display output suitable for low-resolution graphics
3. **Memory expansion solution**
    - Hardware enabling at least 4 MB of pageable memory beyond the native addressing limit.
    - Fully documented memory-mapping and access mechanism.
4. **Design tradeoff analysis**
    - The team will research and justify their choice of hardware solutions.
    - Evaluation of off-the-shelf components versus custom designs.
    - Regular design reviews to justify decisions if AMD funding assistance is used.
    - Power-delivery and interposer design considerations included as needed.

Software Deliverables

1. **DOOM-Inspired Demonstration**
    - Ideally, a highly simplified DOOM-style experience running natively on the Am9080 instruction set.
    - Minimum performance target: a playable demonstration; higher frame rates (e.g., ~15 fps) are aspirational and expected primarily in turbo mode.
    - Grayscale graphics and reduced resolution are acceptable.
    - Verticality and advanced DOOM features may be approximated or omitted.
    - Minimum viable deliverable: a custom renderer and gameplay loop that visually and mechanically resembles classic DOOM-style gameplay.
    - The software must run on:
        - The Am9080 FPGA implementation
        - The Am9080 FPGA implementation in turbo mode

Stretch Goal

If time and resources permit, the team may explore:

- Additional rendering modes (e.g., color or higher resolution)
- Alternative rendering techniques
- Performance experiments enabled by turbo mode

These stretch goals are explicitly **non-blocking** and will not be used as primary success criteria.

Other Information

- Running a DOOM-inspired application on an Am9080-class processor has not been previously demonstrated.
- Similar demonstrations have been achieved on architectures such as the Z80 and 6502 using heavily optimized software and simplified rendering models.
- The team is expected to evaluate feasibility continuously and document limitations and tradeoffs.
- A backup software-only project is defined should hardware integration risks prevent completion of the full system.

Administrative Information

Primary Contacts:

- Rex McCrary (Primary Sponsor)
- Michael Castiglia
- Jerrett Wendt

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